Code-Layout, Readability and Reusability

Date	28 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Followed Python PEP 8 formatting standards
2	Proper File Structure	Files and folders are logically organize	Yes	Separate folders for templates, static files, dataset, and model
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Employed specific identifiers like nitrogen, temperature, humidity, and prediction_result to enhance clarity.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like `predict_crop()`, `load_model()`, and `preprocess_data()` improve readability
5	Code Comments	Inline and block comments explain important logic	Yes	Comments added for preprocessing, model loading, and prediction logic
6	Modular Design	Code is split into reusable functions/modules	Yes	ML model, preprocessing, and Flask routes are implemented as separate modules
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable functions used to avoid duplication
8	Error Handling	Exceptions and errors are handled properly	Yes	Input validation and exception handling implemented for prediction requests

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)

1	Data Preprocessing Module	Python, Pandas	Model training and Prediction	High
2	Machine Learning Model	Sckit-learn	Flask Prediction module	High
3	Prediction Function	Python	Crop Recommendation endpoint	High
4	Flask Routes	Flask	Handle user requests and responses	Medium
5	HTML Templates	HTML, Bootstrap	Input and result pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules and folders
Readability	5	Meaningful naming conventions and consistent formatting improve readability
Reusability	5	Modular functions and reusable components reduce code duplication
Documentation / Comments	4	Important sections are documented with comments; additional documentation can be added if required
Overall Score	4.8	Clean, maintainable, and reusable codebase following good software engineering practices